# Janit Rajkarnikar

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## Education

**University of Southern Mississippi** — Hattiesburg, MS
*Bachelor of Science in Computer Science, Minor in Mathematics and Data Analysis* — *May 2026 (Expected)*
GPA: 4.0

## Work Experience

**Software Engineer Intern** — Aug 2024 – Dec 2024
*Woafmeow, Inc* — *Remote*

- Developed and integrated RESTful APIs in a FastAPI backend with an Expo frontend, ensuring cohesive full-stack performance.
- Designed and implemented secure user authentication and authorization using JWT, enabling scalable access control.
- Led the successful deployment of the application to app stores, managing platform-specific configurations for iOS and Android.

## Projects

**The Designer's Touch** — *React, Node.js, Express, Three.js, Firebase, AWS, Azure*

- Developed a web platform using React, Node.js, and Three.js, enabling users to create personalized merchandise with interactive, real-time 3D visualization.
- Deployed the platform using AWS and Azure, implementing continuous integration and delivery pipelines while maintaining server reliability.
- Built a secure and user-friendly interface with React, Firebase, and Express, allowing users to log in, create and customize artwork, apply it to different sections of apparel, save their creations, leverage AI-generated design suggestions, and complete purchases through a robust checkout system.

**Emotion Tracker (Emoki)** — *React Native, Node.js, Flask, MongoDB, Firebase*

- Implemented real-time emotion detection using RoBERTa models, providing insights into emotional trends and logging intense emotions along with location data for detailed analysis of triggers.
- Designed a chatbot feature that analyzes user messages, tracks emotions, and maintains a personalized diary to help users reflect and manage emotional well-being.
- Optimized the backend with Flask and Express for seamless data processing and integrated Firebase for secure authentication and real-time data synchronization.

**AFM Image Processing Application** — *C#, .NET, WPF, Python*

- Developed a desktop application using .NET 8.0 (WPF) and pythonnet to seamlessly integrate Python-based image processing of AFM .ibw files—no external server required.
- Created robust Python scripts to fully automate AFM .ibw file processing, incorporating data extraction via Igor, advanced statistical modeling with scikit-learn for plane fitting, polynomial leveling, and masking large features using NumPy arrays.
- Leveraged matplotlib for high-quality image generation and grid layouts, ensuring a fully integrated, reproducible, and efficient pipeline for transforming raw microscopy data into ready-to-use, visually informative results.

**Calorie Tracker (CalTrack)** — *React Native, SQLite, Node.js*

- Developed a cross-platform mobile app using React Native to monitor daily calorie intake, featuring an intuitive user interface and real-time goal-setting with tracking and reminders to help users achieve health objectives.
- Optimized SQLite for efficient storage and retrieval of food logs, integrating the USDA FoodData Central API for accurate nutrient analysis and providing users with actionable dietary insights.

## Technical Skills

Programming Languages: JavaScript, C++, Python, C#, Visual Basic

Libraries/Frameworks: Node.js, Django, React, Next.js, Three.js, React Native, Flask, PyTorch, .NET, Matplotlib, pandas, Unity

Tools/Platforms: Git, Firebase, AWS, Azure, Expo, WPF

Databases: SQLite, MongoDB, PostgreSQL